

US Accidents (2016-2023): Statistical Analysis — Executive Summary

Prepared as a Data Analyst/BI portfolio project. Full technical detail is in `notebooks/01_eda_and_hypothesis_testing.ipynb`; this document is the non-technical summary of what was found and why it matters.

Project overview

Statistical analysis of ~7.7 million US traffic accident records (2016-2023), examining what factors are associated with accident severity — weather, time of day, and road features — using hypothesis testing and logistic regression, presented alongside an interactive dashboard.

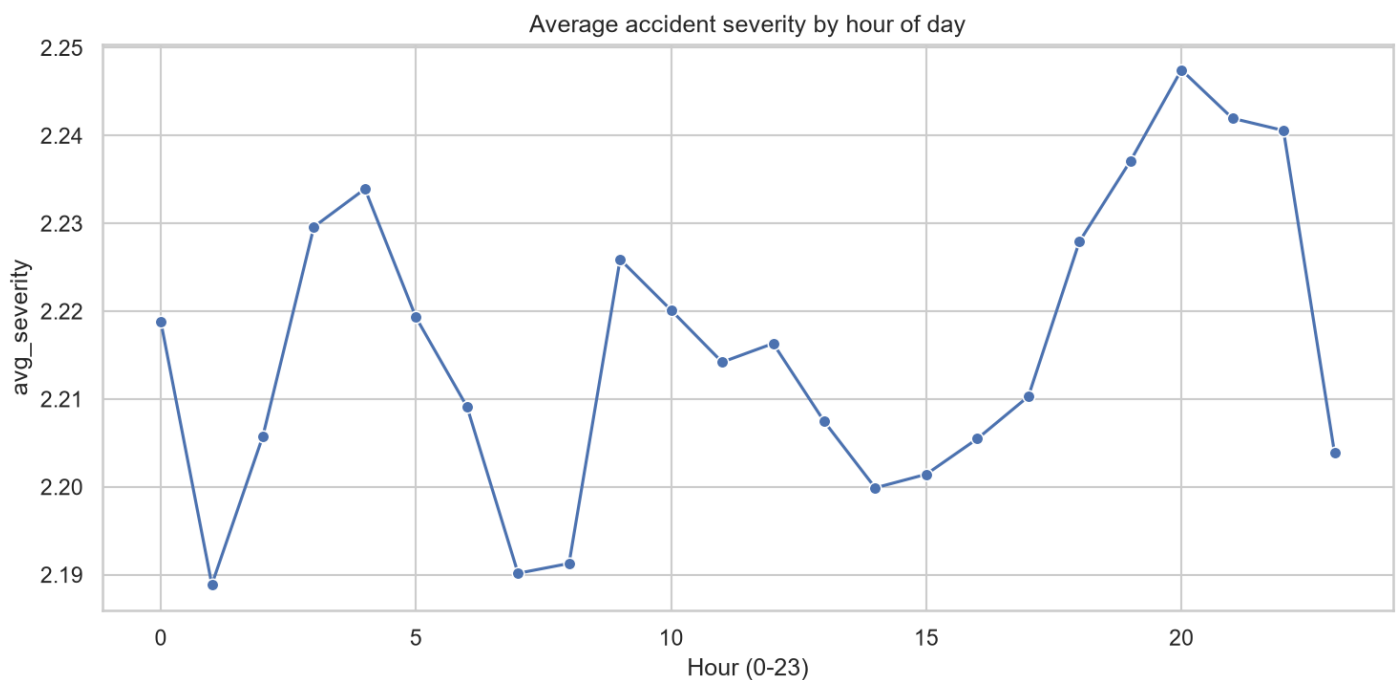
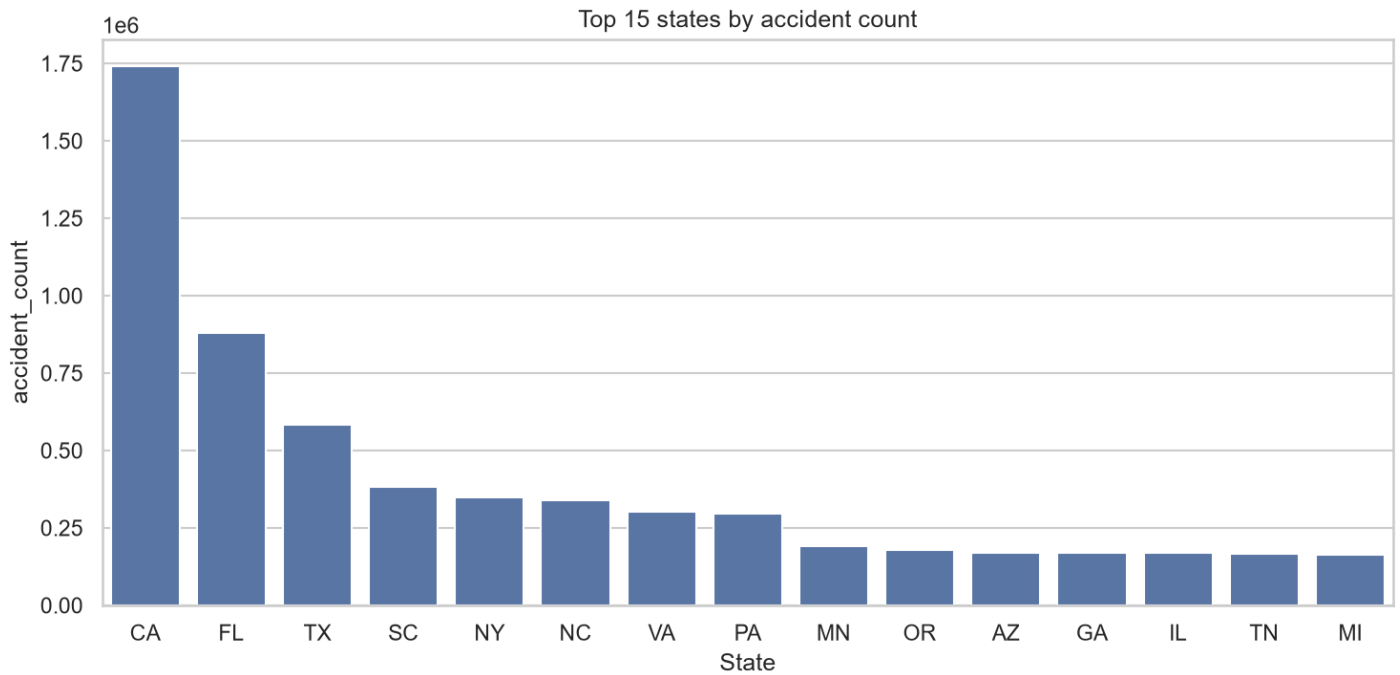
Findings

Based on a reproducible 200,000-accident sample drawn from the full 7.7-million-record dataset:

- **Weather matters.** Accident severity is strongly associated with weather conditions at the time of the crash — this isn't a small or coincidental pattern; it's one of the strongest relationships found in the data.
- **Time of day matters.** Average accident severity meaningfully shifts depending on the hour of day the accident occurred.
- **Road features are strong, reliable predictors of severity.** Accidents near a **junction** are meaningfully more likely to be high-severity than accidents without one nearby. Conversely, accidents near a **crossing**, a **traffic signal**, or a **stop sign** are all meaningfully *less* likely to be high-severity — consistent with these features generally slowing traffic down before a potential collision. All four effects are statistically robust, not noise.

Full statistical detail (test statistics, p-values, confidence intervals, odds ratios) is in `notebooks/01_eda_and_hypothesis_testing.ipynb` and `vault/Findings.md`.

Key visualizations



Dashboard

An interactive Streamlit dashboard covers:

- A **US map** of accident counts by state, plus a top-20 state bar chart.
- A **year-over-year trend** of accident counts, 2016-2023.
- **Cross-filterable** weather-condition and hour-of-day breakdowns — pick a state and/or year and both charts update to that slice of the data.
- A **key drivers panel** surfacing the logistic regression findings (odds ratios with confidence intervals) directly in the dashboard, not just the notebook.

Run locally with `uv run streamlit run dashboard/app.py`, or see the live link below.

Live demo

[Live dashboard](#)

Note: on Render's free tier the service sleeps after inactivity and may take ~30 seconds to wake on first load.

Conclusion

This project demonstrates an end-to-end Data Analyst/BI workflow: scalable data processing (DuckDB) on a real 7.7M-record dataset, rigorous statistical inference (hypothesis testing, logistic regression with interpreted effect sizes) rather than purely descriptive analysis, and a deployed, interactive way to explore the results — with full reproducibility via automated tests and CI/CD.